Medicaid Expansion and Out-of-Pocket Costs: Causal Evidence from Double Machine Learning

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Section One: Introduction

Access to affordable health insurance is crucial for economic stability and public health. Before the implementation of the Affordable Care Act (ACA), millions of low-income adults faced significant barriers in accessing health insurance due to unaffordable premiums, restrictive Medicaid eligibility, and pre-existing condition exclusions. The ACA's Medicaid expansion, initiated in 2014, targeted adults earning up to 138% of the federal poverty level (FPL), creating a natural experiment to evaluate how policy interventions can alter insurance accessibility, affordability, and associated financial burdens.

This study aims to investigate the causal relationship between insurance coverage resulting from the ACA Medicaid expansion and medical out-of-pocket (MOOP) expenses. Employing advanced causal inference methods, specifically, Double Machine Learning Instrumental Variable (DML-IV) analysis, we explore this relationship across different socioeconomic subgroups defined by income levels (low-FPL and mid-FPL). Our empirical strategy leverages quasi-experimental variations between Medicaid expansion and non-expansion states, utilizing Difference-in-Differences (DiD) frameworks supplemented by rigorous machine learning algorithms (Random Forest, LASSO, and XGBoost) for robustness.

By integrating Grossman's (1972) health capital model, which posits that insurance reduces the marginal cost of health investments, and Arrow's (1963) theory of market failures in healthcare, this study contextualizes the economic mechanisms through which Medicaid expansion influences health investment decisions and subsequent financial outcomes. The findings from this analysis offer valuable insights for policymakers concerning the effectiveness of Medicaid expansions, particularly regarding financial protections and insurance market dynamics.

Section Two: Literature review

The evidence base on Medicaid expansion uniformly shows sizable gains in insurance coverage and meaningful financial relief for low-income adults. Sommers et al. (2014) delivered the first national estimates, documenting sharp drops in the uninsurance rate—especially in states that adopted the ACA expansion. Using stronger quasi-experimental designs, Courtemanche et al. (2017, 2019, 2020) confirmed and extended these findings: expansion states posted nearly twice the coverage gains of non-expansion states, and the improvements persisted through 2018 while narrowing racial and socioeconomic gaps using a DDD framework.

Beyond coverage, researchers have traced clear downstream effects on household finances and health-care behaviour. The Oregon Health Insurance Experiment (Finkelstein et al., 2012) showed that new Medicaid enrollees faced substantially lower out-of-pocket (MOOP) spending. Simon et al. (2016) employed a Seemingly Unrelated Regression (SUR) framework within a DiD design to test whether expansion moved several correlated outcomes simultaneously; they found larger use of preventive services with no evidence of moral hazard. By contrast, Courtemanche et al. (2018) detected concurrent increases in both preventive care and some risky behaviours, highlighting behavioural complexity in response to expanded coverage.

Building on this literature, our study integrates modern machine-learning tools with a Double-Machine-Learning IV estimator to rigorously adjust for high-dimensional confounders. This approach lets us ask not only whether Medicaid expansion raised coverage, but whether it produced coordinated improvements in OOP spending across distinct income subgroups—evidence crucial for the remaining hold-out states and for tailoring complementary policies that maximize the financial benefits of insurance.

Section Three: Economic Theories and Hypothesis

The theoretical foundation for this study is rooted in Grossman's (1972) health capital model and Arrow's (1963) market failure theory. According to Grossman, health can be viewed as a form of capital stock that depreciates over time but can be maintained or improved through investments in healthcare services. Insurance reduces the marginal cost of healthcare investments, thereby encouraging more frequent and timely medical care utilization, potentially leading to better health outcomes and lower future healthcare expenses.

Grossman's health capital model explicitly provides the economic mechanism through which Medicaid expansion operates, defining the marginal cost of health investments (π) as:

$$\pi = \frac{P_M}{\partial I/\partial T} + \frac{w}{\partial I/\partial M}$$

Where:

- π is the marginal cost of health investments.
- P_M is the price of medical care (e.g., OOP cost and premium); w is the wage rate.
- $\partial I/\partial T$ is the marginal productivity of time in health production.
- $\partial I/\partial M$ is the marginal productivity of medical care in improving health.

The ACA Medicaid expansion significantly reduces individuals' costs by decreasing out-of-pocket expenses, resulting in a net reduction in medical care prices (P_M) for enrollees. Lower P_M reduces the marginal cost of health investments (π), making healthcare investments more attractive and incentivizing insurance enrollment. Increased enrollment further dilutes risk within the insurance pool, potentially lowering premiums.

Arrow's market failure theory complements this framework by highlighting inefficiencies such as moral hazard, adverse selection, and information asymmetry prevalent in healthcare markets. The ACA Medicaid expansion specifically addresses these inefficiencies by reducing financial barriers to healthcare access, thereby potentially correcting distorted healthcare market pricing.

Given the instrumental variable (IV) approach utilized in this study, we formulate the following hypothesis explicitly reflecting this causal pathway:

Hypothesis: ACA Medicaid expansion increases insurance coverage, which in turn causally decreases medical out-of-pocket expenses for enrolled individuals.

While Grossman's model provides valuable insights into health behaviors, it relies on simplifying assumptions such as exogenous depreciation rates and absence of credit constraints, and ignores behavioral factors like bounded rationality and uncertainty in health shocks or investment returns. Despite these limitations, it remains a cornerstone model, effectively illustrating why individuals invest resources into health even without immediate illness.

Section Four: Methodology

Data Sources and Sample Selection

We draw on individual‑level microdata from the March ASEC of the CPS—a nationally representative survey widely used in labor and health economics conducted every March—and augment it with state‑level administrative data. Our analytic sample comprises non‑elderly adults aged 20–65 at the time of interview (which, after subtracting one year to align with the insurance‑coverage window, corresponds to ages 19–64). To assess pre‑trend validity and avoid potential instability associated with the early Trump administration, we pool CPS ASEC data from 2011 through 2017. Throughout, we apply the person‑level sampling weights (scaled to recover annual estimates) to preserve national representativeness.

We further restrict the sample by excluding individuals with missing values on key variables and drop proxy respondents and those in group quarters to ensure measurement consistency and comparability across states. After these restrictions, our final sample includes approximately *786,258* observations across the states.

To account for macroeconomic variation across time and space—particularly differences in labor market recovery trajectories following the 2008–2009 financial crisis—we merge annual, seasonally adjusted state unemployment rates from the Bureau of Labor Statistics' Local Area Unemployment Statistics (LAUS). In addition, we incorporate information on state Medicaid expansion decisions and effective implementation dates from the Kaiser Family Foundation (KFF), cross-validated against timelines published in Courtemanche et al. (2017) and Simon (2016). These sources are widely recognized for their accuracy and reliability in health policy research.

While the CPS ASEC offers rich individual‑level data, it is still subject to some limitations. First, insurance coverage variables are self‑reported and refer to respondents' status one year prior to the survey, which can introduce recall error or misclassification—particularly for Medicaid, which is often underreported. To address this temporal mismatch, we subtract one year from both age and state of residence so that these variables align with the coverage window. Unfortunately, other covariates—marital status, household size, unemployment status, and educational attainment—cannot be similarly lagged, and we assume they remain unchanged over the recall period; we believe any resulting bias to be minimal. Second, the cross-sectional design of the CPS limits our ability to track individuals over time, although state-year variation supports a credible DiD framework. Third, ACA implementation timing may vary within states or overlap with other state-level policies not fully captured in our data; we mitigate this by including state and year fixed effects and by validating expansion dates using multiple sources. Fourth, small sample sizes in late-expanding states (e.g., Montana, Alaska) could reduce power to detect heterogeneous effects.

Model Specification

We estimate the causal impact of insurance on medical out-of-pocket (MOOP) spending using the partial-linear instrumental-variables estimator with Double/Debiased Machine Learning (PLIV-DML) developed by Chernozhukov et al. (2018). In essence, the method combines flexible machine-learning prediction with a classical two-stage least-squares core, ensuring that high-dimensional controls do not introduce bias. We instrument insurance coverage with the interaction of expansion status and post-expansion period ($TREATMENT_S \times POST_t$), exploiting exogenous variation from the ACA Medicaid expansion. By constructing Neyman-orthogonal moment functions and using cross-fitting, PLIV-DML dramatically reduces regularization bias and guards against overfitting, while still allowing any black-box learner (e.g. Random Forest, LASSO, XGBoost) to flexibly model the outcome, treatment, and instrument relationships. Moreover, its orthogonal score construction affords a form of double robustness: as long as either the treatment or the instrument nuisance model is consistently estimated, the causal parameter estimate remains reliable. These features make PLIV-DML especially well suited for policy evaluation in complex, high-dimensional survey data like the CPS ASEC.

Our regressions include state and year fixed effects, the seasonally-adjusted state-level annual unemployment rate, and a rich set of individual covariates (age and state of residence lagged one year, gender, marital status, household size, race/ethnicity and education dummies, citizenship, student status, and continuous household income) to absorb time-invariant heterogeneity and observed confounders. We examine financial burden using medical out-of-pocket (MOOP). MOOP is winsorized at the 1st and 99th percentiles before estimation due to its right-skewed distribution.

We split the sample into three folds. Each fold in turn is held out while the other two are used to train machine‑learning models (Random Forest, LASSO, XGBoost) that predict the outcome, the treatment, and the instrument from the controls. Using those predictions we remove, or residualise, the contribution of the controls from MOOP, insurance, and the instrument for every observation in the held‑out fold.

On these residuals we run a simple two‑stage least‑squares regression: the residualised instrument is the first‑stage predictor for residualised insurance; the second stage regresses residualised MOOP on the fitted insurance values. The resulting coefficient $\hat{\beta}$ is free of regularisation bias because the score is Neyman‑orthogonal.

The entire algorithm is run separately for the low‑FPL (below the poverty line) and mid‑FPL (100–138 percent of poverty) subsamples to uncover heterogeneous effects.

Section Five: Empirical Results

Descriptive statistics

Table F1 provides an overview of the sample characteristics, offering context on insurance coverage, financial outcomes and demographic composition, before and after Medicaid expansion across expansion and non-expansion states. While many covariates show statistically significant differences between expansion and non-expansion states prior to 2014, the magnitude of these differences is generally small. This suggests that selection into treatment is not random, but observable imbalance is limited, supporting the use of covariate adjustment and fixed effects in our regression framework.

Table F1 also presents expansion-states exhibited higher insured rates even before the policy, and saw larger post-2014 increases in coverage, driven primarily by gains in Medicaid enrollment. Individuals in expansion states had slightly lower out-of-pocket medical expenses both before and after expansion. These differences widened post-2014, implying improved financial protection.

These descriptive findings provide preliminary evidence that Medicaid expansion was associated with improved coverage and affordability, though causal claims are addressed through regression analysis in the following sections.

Overview of findings

Table F2 summarises the DML‑IV point estimates $\hat{\beta}$, heteroskedasticity-robust standard errors, 95 % confidence intervals, and predictive‑fit diagnostics for each learner–subgroup combination.

- Random Forest (ranger) delivers consistently negative and statistically significant estimates in both income groups, indicating that gaining insurance reduces annual MOOP by roughly $3,650 for mid‑FPL adults and $1,250 for low‑FPL adults.
- LASSO also yields a sizable and significant negative effect in the mid‑FPL group (about $7,300), but its low‑FPL estimate is small and imprecise, failing to reject the null.

- XGBoost produces large positive but highly imprecise coefficients in both groups; with standard errors dwarfing the point estimates, these results are statistically indistinguishable from zero and signal that the learner struggled to capture a reliable first stage.

Taken together, two of the three learners point to meaningful cost savings from insurance, and no learner finds a statistically significant increase in MOOP, reinforcing the directional robustness of our main conclusion.

Heterogeneity by income group

Mid‑FPL adults (100–138 % poverty) experience the largest and most precisely estimated reductions. Both Random Forest and LASSO estimates are negative and significant at conventional levels ($p < 0.05$), suggesting that near‑poor adults who newly qualify for Medicaid benefit most in absolute dollar terms.

Low‑FPL adults (< 100 % poverty) show a more modest but still significant reduction using the Random Forest learner. The lack of precision in the LASSO and XGBoost runs hints that noise in the first‑stage prediction may obscure the true effect for this extremely low‑income group, but the direction remains either negative or statistically null.

Model performance and diagnostics

Out‑of‑sample MSEs for the outcome (Y‑), treatment (D‑), and instrument (Z‑) models shown in Table F3 confirm that Random Forest attains the best predictive fit, which likely explains its sharper confidence intervals. The low MSE_Z scores across learners further reassure us that the instrument, the Medicaid‑expansion indicator, is being modelled with adequate accuracy, a prerequisite for strong first‑stage relevance.

Section Six: Conclusion

Discussion

In sum, the DML-IV analysis indicates that Medicaid-expansion-induced increases in insurance coverage lowered out-of-pocket medical spending, with the clearest evidence among adults between 100% and 138% of the federal poverty line. Although one learner (XGBoost) fails to produce precise estimates, the convergence of Random Forest and LASSO results — together with rigorous out-of-sample diagnostics confirming adequate first-stage fit — supports the conclusion that insurance coverage causally reduces MOOP spending.

Policy implications

Expansion hold-outs. The sizable cost savings we document strengthen the economic case for expansion in the remaining non-expansion states. By shielding families from unpredictable and potentially catastrophic bills, Medicaid can enhance financial stability and reduce reliance on informal debt.

Benefit design. The larger absolute savings for mid-FPL adults suggest that modest cost-sharing waivers still leave meaningful exposure for households just above the poverty line. Policymakers could consider sliding-scale reductions in copayments and deductibles to ensure the full protective value of coverage.

Targeted outreach. Because the poorest adults benefited less in dollar terms — partly reflecting lower pre-expansion spending — efforts to improve benefit take-up (e.g., simplified enrollment, automatic renewal) remain critical for maximising equity.

Limitations

Several caveats temper the interpretation of our findings. First, the identifying assumption that state Medicaid expansion is conditionally exogenous though cannot be proven, so residual policy endogeneity may bias the estimates. Second, MOOP is self-reported in CPS-ASEC and thus susceptible to recall error. Third, the cross-sectional design captures snapshots rather than longitudinal coverage spells, leaving open questions about persistence. Fourth, practical constraints of the current DoubleML implementation meant that survey weights were omitted and standard errors were heteroskedasticity-robust but not clustered at the state level; if sampling weights or within-state correlation are important, the point estimates and their precision could be modestly biased. Finally, to keep computation feasible we employed only three-fold cross-fitting, which preserves consistency but may leave some finite-sample variance on the table compared with a greater number of folds.

Despite these limitations, this study demonstrates the promise of combining machine-learning nuisance estimation with instrumental-variables identification for policy evaluation. The consistent finding across multiple learners that Medicaid expansion reduced out-of-pocket spending reinforces the broader evidence that public insurance expansions provide meaningful financial protection for low-income populations. Future work could extend the analysis by incorporating survey weights into the DML framework, increasing the number of cross-fitting folds, and examining additional outcome domains such as medical debt, health-care utilization, and subjective financial well-being.

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Appendix

Appendix A: Treatment Variable Construction

Following established literature (e.g., Courtemanche et al., 2017; Simon, 2016), we define a binary treatment variable, $Treatment_s$, which equals 1 if the states had implemented the ACA Medicaid expansion before 2017, and 0 otherwise. It is noteworthy that some states that adopted the expansion after the federal 2014 baseline are late adopters, with implementation dates as follows: Michigan (April 2014), New Hampshire (August 2014), Pennsylvania (January 2015), Indiana (February 2015), Alaska (September 2015), Montana (January 2016), and Louisiana (July 2016). The following Table A1 is the full list of expansion states ($Treatment_s$=1)and non-expansion states($Treatment_s$=0). We also define a binary treatment variable, $Post_t$, which equals 1 if the state had implemented the ACA Medicaid expansion as of survey year t, regardless of the month that took effect, and 0 otherwise.

Table A1: Classification of Expansion and Non-Expansion States

Expansion States (2014–2018)	**Non-Expansion States**
Arizona	Alabama
Arkansas	Florida
California	Georgia
Colorado	Idaho
Connecticut	Kansas
Delaware	Maine
District of Columbia	Mississippi
Hawaii	Missouri
Illinois	Nebraska
Iowa	North Carolina
Kentucky	Oklahoma
Maryland	South Carolina
Massachusetts	South Dakota
Minnesota	Tennessee
Nevada	Texas
New Jersey	Utah
New Mexico	Virginia
New York	Wisconsin
North Dakota	Wyoming
Ohio	
Oregon	
Rhode Island	
Vermont	
Washington	
West Virginia	
Michigan (Apr 2014)	
New Hampshire (Aug 2014)	
Pennsylvania (Jan 2015)	
Indiana (Feb 2015)	
Alaska (Sep 2015)	
Montana (Jan 2016)	
Louisiana (Jul 2016)	

Notes: This table shows the state classification as regards Medicaid eligibility for adults, including 50 states as well as the District of Columbia.

Appendix B: Outcome Variable Construction

We examine four main categories of outcomes:

1. **Insurance coverage**: measured through binary indicators for any coverage ($Insured$).

2. **Financial burden**: proxied by total out-of-pocket medical spending, which is the total amount the family actually paid for medical services ($MOOP$) MOOP is winsorized at the 1st and 99th percentiles before estimation.

Appendix C: Subgroup construction

As a measure of socioeconomic status, we calculate each household's income-to-poverty ratio based on the Department of Health and Human Services' Federal Poverty Guidelines, which adjust for household size and year. For stratified analyses, we use the following standard poverty thresholds:

- Low-FPL (< 100%):
  - In expansion states, individuals become eligible for Medicaid.
  - In non-expansion states, individuals remain ineligible for both Medicaid and subsidized exchange coverage (the "coverage gap").
- Mid-FPL (100 – 138%):
  - In expansion states, individuals gain Medicaid coverage.
  - In non-expansion states, individuals qualify for subsidized exchange plans (e.g., premium tax credits on ACA marketplaces).

We also treat the income-to-poverty ratio as a continuous covariate in our regressions to capture fine-grained variation. Ultimately, we define two analysis subgroups:

1. $Mid - FPL$ (100–138%)
2. $Low - FPL$(below 100%)

Appendix D: Control Variables construction

We control for a rich set of individual‑level characteristics: age and state of residence (both lagged by one year to align with insurance questions about "one year ago"), gender, marital status, household size, and current unemployment status. We include race/ethnicity indicators for Hispanic, non-Hispanic Black, Asian, and Other (with non-Hispanic White as the omitted category), and educational attainment dummies for high-school graduate, some college, and college graduate (below high school is the reference group). Continuous household income is also included. To capture local labor-market conditions, we add the seasonally adjusted, state-level annual unemployment rate. Finally, state and year fixed effects absorb time-invariant regional heterogeneity and common national shocks.

Because insurance coverage is reported with a one-year lag, we adjust both age and state of residence accordingly; unfortunately we cannot similarly lag marital status, household size, unemployment status, or educational attainment, so we assume these remain unchanged over the recall period and believe any resulting bias to be minimal. These covariates follow the specifications in Courtemanche et al. (2017) and Simon et al. (2016).

Appendix E: Empirical Strategy

To estimate the causal effect of insurance coverage on out-of-pocket (MOOP) spending, we implement the **Partial-Linear Instrumental-Variables estimator with Double/Debiased Machine Learning (PLIV-DML)** of Chernozhukov et al. (2018). This approach flexibly adjusts for high-dimensional covariates without incurring overfitting bias and yields Neyman-orthogonal moment conditions for robust inference.

Data and Instrument

- Outcome: $Y_{ist} = MOOP$
- Treatment: $D_{ist} = INSURED$
- Instrument: $Z_{ist} = TREATMENT_S \times POST_t$ where $TREATMENT_S$ indicates ever-expansion and $POST_t$ indicates survey year ≥ implementation.
- Controls: rich vector X_{ist} including lagged age, lagged state, gender, marital status, household size, race/ethnicity, education dummies, citizenship, student status, household income, state unemployment rate, and state & year fixed effects.

Cross-Fitting and Residualization

- Randomly split the sample into $K = 3$ folds.
- For each fold K:
  1. Train three machine-learning regressors (Random Forest, LASSO, XGBoost) on the other $K - 1$ folds to predict:
     a. $\hat{u}(X) \approx E[Y|X]$
     b. $\hat{v}(X) \approx E[D|X]$
     c. $\hat{\pi}(X) \approx E[Z|X]$
  2. For observations in fold k, compute residuals:

     $\tilde{Y} = Y - \hat{u}(X),\ \tilde{D} = D - \hat{v}(X),\ \tilde{Z} = Z - \hat{\pi}(X)$

Neyman-Orthogonal IV Estimation

- Pool residuals ($\widetilde{Y}, \widetilde{D}, \widetilde{Z}$) across all folds and run the two-stage least squares:

  First stage: $\widetilde{D} = \alpha_1 + \theta_1 \widetilde{Z} + u$, Second stage: $\widetilde{Y} = \alpha_2 + \hat{\beta}\widetilde{D} + v$

- The estimator $\hat{\beta}$ is orthogonal to first-stage regularization bias, yielding valid inference.

Subgroup Analysis

- The entire algorithm is applied separately to two subsamples:
  - $Mid - FPL$ (100–138%)
  - $Low - FPL$ (below 100%)
- This uncovers heterogeneous treatment effects by income group.

Inference and Robustness

- Standard errors computed via the built-in DML bootstrap with sample splitting.
- We report point estimates $\hat{\beta}$, heteroskedasticity-robust SEs, 95 % CIs, and p-values.
- Predictive diagnostics: out-of-sample MSE for each of the three nuisance residuals ($\widetilde{Y}, \widetilde{D}, \widetilde{Z}$) to assess first-stage fit.

Appendix F: Table and figures

Tables

Table F1 Descriptive statistics for demographic variables

	Expansion states		Non-expansion states		Pre-2014
Variable	2011–13	2014–17	2011–13	2014–17	difference
Age	41.18 (12.68)	41.24 (12.80)	41.03 (12.62)	41.22 (12.75)	0.15***
Household Income	91,116 (94,739)	103,056 (106,934)	80,387 (83,350)	91,195 (97,526)	10,729***
Female	0.52 (0.50)	0.52 (0.50)	0.52 (0.50)	0.52 (0.50)	−0.00
Married	0.57 (0.49)	0.57 (0.50)	0.60 (0.49)	0.59 (0.49)	−0.03***
Family Size	3.18 (1.63)	3.18 (1.64)	3.14 (1.58)	3.15 (1.58)	0.04***
Race					
Hispanic	0.17 (0.38)	0.20 (0.40)	0.17 (0.38)	0.19 (0.39)	−0.00
Black	0.10 (0.30)	0.10 (0.30)	0.14 (0.35)	0.15 (0.36)	−0.04***
Asian	0.07 (0.26)	0.08 (0.28)	0.03 (0.17)	0.03 (0.18)	0.04***
Other Race	0.03 (0.18)	0.04 (0.19)	0.03 (0.17)	0.03 (0.16)	0.01***
Education / Citizenship					
HS Grad	0.28 (0.45)	0.28 (0.45)	0.29 (0.46)	0.29 (0.45)	−0.01***
Some College	0.29 (0.46)	0.29 (0.45)	0.31 (0.46)	0.31 (0.46)	−0.02***
College Grad	0.32 (0.47)	0.34 (0.47)	0.28 (0.45)	0.30 (0.46)	0.04***
Foreign Born	0.20 (0.40)	0.22 (0.41)	0.16 (0.37)	0.17 (0.38)	0.04***
U.S. Citizen	0.90 (0.30)	0.89 (0.31)	0.91 (0.29)	0.91 (0.29)	−0.01***
Student	0.07 (0.25)	0.07 (0.26)	0.06 (0.24)	0.07 (0.25)	0.01***
Insurance & Cost					
Insured	0.83 (0.38)	0.90 (0.30)	0.78 (0.41)	0.84 (0.36)	0.05***
MOOP	3,890 (4,792)	4,447 (5,332)	3,992 (4,746)	4,618 (5,371)	−102***

Notes: Means with standard deviations in parentheses. Stars denote significance of the 2011–13 expansion vs. non-expansion difference: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table F2 DML Estimate of the Effect of Insurance on Out-of-Pocket Spending

Learner	Subgroup	$\hat{\beta}$	SE	95% CI	p-value
ranger	MIDFPL	−3652.03	1051.31	[−5712.56 −1591.49]	0.0005
ranger	LOWFPL	−1251.83	624.29	[−2475.41 −28.25]	0.0449
lasso	MIDFPL	−7319.08	3070.02	[−13336.21 −1301.96]	0.0171
lasso	LOWFPL	−360.16	887.16	[−2098.96 1378.64]	0.6848
xgboost	MIDFPL	10886.49	20662.81	[−29611.87 51384.86]	0.5983
xgboost	LOWFPL	18727.56	46952.98	[−73298.60 110753.71]	0.6900

Table F3 Cross-Validation MSEs for Nuisance-Function Fits

Learner	Subgroup	MSE_Y	MSE_D	MSE_Z
ranger	MIDFPL	11505597	0.15726	0.0043571
ranger	LOWFPL	8388336	0.17242	0.0028977
lasso	MIDFPL	12542638	0.17297	0.0637270
lasso	LOWFPL	9094137	0.18791	0.0637983
xgboost	MIDFPL	11036313	0.15866	0.0007338
xgboost	LOWFPL	8313868	0.17147	0.0007435

Appendix G: Software and Reproducibility

Data processing and initial weighting checks are carried out in R 4.3 .

Appendix H: Use of AI Assistance

This paper was drafted with structural and editorial assistance from OpenAI's ChatGPT. All substantive modeling decisions, data construction, and estimation commands were conceived and executed by the authors.